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NTERNSHIP DOMAIN: AI AUTOMATION

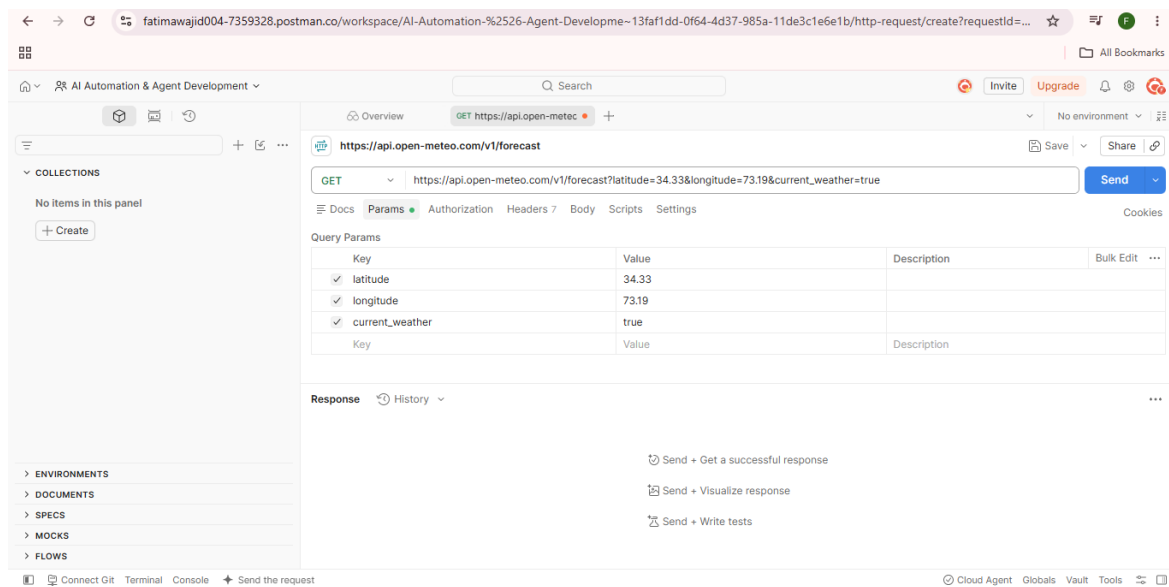
Day 7 – API Fundamentals & Postman: API Exploration Report

API: Open-Meteo Forecast API

Request Method: GET

Endpoint: <https://api.open-meteo.com/v1/forecast>

Query Parameters: latitude=34.33, longitude=73.19, current_weather=true



200 Success:

Full JSON Response

```
{
  "latitude": 34.340946,
  "longitude": 73.15327,
  "generationtime_ms": 7.50422477722168,
  "utc_offset_seconds": 0,
  "timezone": "GMT",
  "timezone_abbreviation": "GMT",
  "elevation": 1072.0,
  "current_weather_units": {
    "time": "iso8601",
    "interval": "seconds",
    "temperature": "°C",
    "windspeed": "km/h",
    "winddirection": "°"
  }
}
```

```

    "is_day": "",
    "weathercode": "wmo code"
  },
  "current_weather": {
    "time": "2026-08-05T09:45",
    "interval": 900,
    "temperature": 28.9,
    "windspeed": 4.0,
    "winddirection": 275,
    "is_day": 1,
    "weathercode": 51
  }
}

```

Json Structure

1. Location Metadata

This section provides information about the requested location and response generation.

- latitude – Latitude of the requested location.
- longitude – Longitude of the requested location.
- generationtime_ms – Time taken by the API to generate the response.
- utc_offset_seconds – UTC offset for the location.
- timezone – Time zone of the location.
- timezone_abbreviation – Short form of the time zone.
- elevation – Elevation of the location in meters.

2. current_weather_units

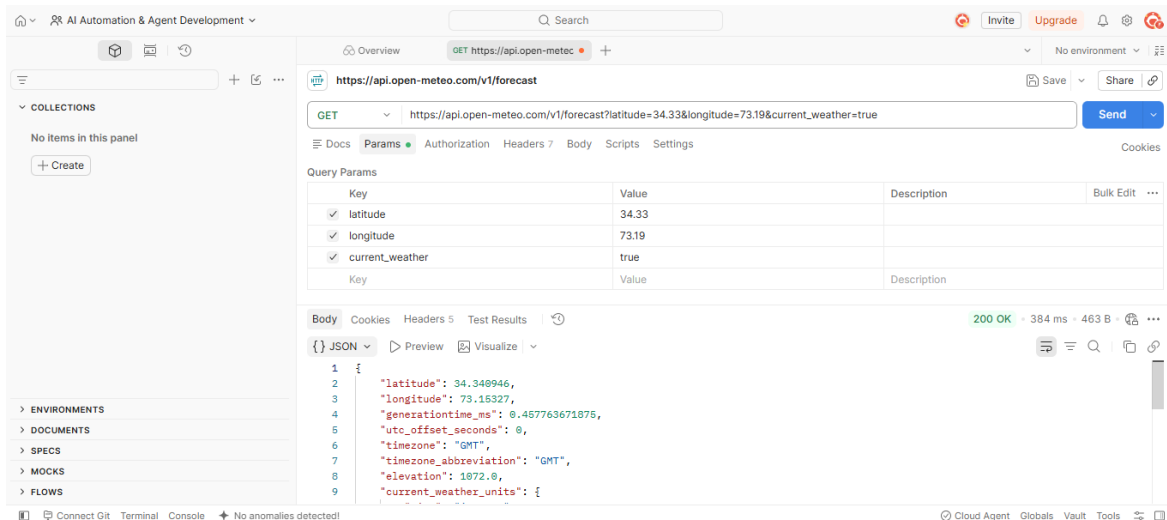
This object defines the measurement units used for each weather field.

- temperature → Degrees Celsius (°C)
- windspeed → Kilometers per hour (km/h)
- winddirection → Degrees (°)
- time → ISO 8601 date-time format
- weathercode → WMO weather code

3. current_weather

This object contains the actual live weather information.

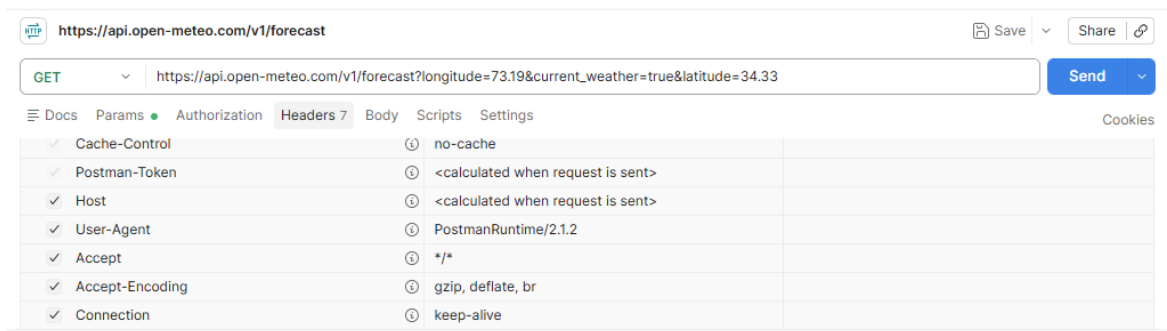
- time – Timestamp of the weather observation.
- interval – Time interval between updates.
- temperature – Current air temperature.
- windspeed – Current wind speed.
- winddirection – Wind direction in degrees.
- is_day – Indicates whether it is daytime (1) or nighttime (0).
- weathercode – Weather condition represented using the WMO weather code.



Header:

Headers: Accept: */*, User-Agent: PostmanRuntime/2.1.2, Accept-Encoding: gzip, deflate, br,

Connection: keep-alive, Cache-Control: no-cache, Host: api.open-meteo.com



Error Test (400 Bad Request)

I intentionally removed the **latitude** query parameter while keeping longitude. The API returned **HTTP 400 Bad Request** with the message: "Parameter 'latitude' and 'longitude' must have the same number of elements". This confirms that both latitude and longitude are required together for a valid request.

The screenshot shows a REST client interface with the following details:

- URL:** `https://api.open-meteo.com/v1/forecast`
- Method:** GET
- Query Params:**

Key	Value	Description
longitude	73.19	
current_weather	true	
- Status:** 400 Bad Request (380 ms, 273 B)
- Body (JSON):**

```
{  "error": true,  "reason": "Parameter 'latitude' and 'longitude' must have the same number of elements"}
```